

Git & GitHub Interactive Workshop

Modern hands-on training for beginners

Agenda

- Git fundamentals
- Repositories & commits
- GitHub collaboration
- Branches & PRs
- Quiz & challenge

Version Control Problem

- Many file versions
- Lost work
- Difficult collaboration
- No audit trail

Git Architecture

Working Directory

Staging Area

Repository

Activity 1: Initialize Git

- `mkdir git-workshop`
- `cd git-workshop`
- `git init`
- `git status`

Activity 2: Track Changes

- Create file
- `git status`
- `git add .`
- Observe staging

Activity 3: Commit Changes

- `git commit -m`
- Meaningful messages
- `git log`
- Commit history

Git Workflow Graphic



GitHub Basics

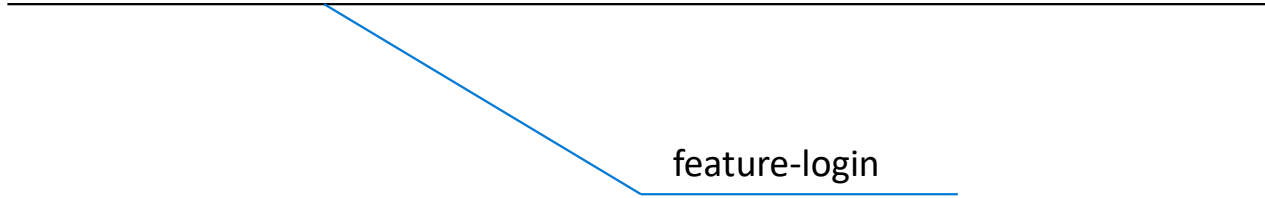
- Remote repositories
- Backup code
- Collaboration
- Portfolio

Push & Pull

- `git remote add origin URL`
- `git push origin main`
- `git pull origin main`
- Sync changes

Branching Illustration

main



Merge Concepts

- Combine branch changes
- Keep main stable
- Resolve conflicts
- Review before merge

Pull Requests

- Code reviews
- Comments
- Approvals
- Merge after review

Interactive Exercise

- Create branch
- Add feature
- Commit
- Merge back

```
mkdir challenge
cd challenge
git init
echo "Git Workshop" > readme.txt
git add .
git commit -m "Initial commit"
git checkout -b feature-profile
echo "Profile Page" >> readme.txt
git add .
git commit -m "Added profile page"
git checkout main
git merge feature-profile
```

 **Git tracks history. GitHub shares history.**
Branches protect history. Pull Requests
improve history.

Quiz 1

- Which command initializes a repo?
- A add B init C push D merge

Quiz 2

- What does git commit create?
- A Snapshot B Folder C Branch D Server

Quiz 3

- Which command uploads code?
- A pull B push C log D status

Best Practices

- Commit often
- Use branches
- Review code
- Write clear messages

Hands-On Challenge

- Initialize repo
- Create branch
- Commit updates
- Merge changes

Q&A and Resources

- Git docs
- GitHub Skills
- Practice daily
- Thank you